



National University of Modern Languages

Lab Task

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Class:	BSCS VII-A
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Task

Objective:

Create a basic 2D car dashboard with a keyboard. Use simple non-parameterized animation to show a key press (like changing color when a key is pressed).

Task Instructions:

1. Create a Simple Car Dashboard Scene (Top View):

- Use rectangles to draw:
 - Dashboard
 - A basic steering wheel (just a circle)
 - A keyboard with 5 rectangular keys placed horizontally

2. Add Non-Parameterized Animation:

- When the user presses a specific key on the actual keyboard (e.g., 'A', 'B', 'C', 'D', 'E'):
 - Change the color of the corresponding key on the screen (e.g., from gray to green for 1 second)

Use `glutKeyboardFunc()` or a similar function for key detection

Use a fixed duration for the color change (hardcoded logic, no parameters).

Bonus (Optional for Extra Marks):

Add a small screen above the keyboard that displays a static message like “Welcome” or “Car System Active”

Learning Outcomes:

- Practice drawing shapes using OpenGL or a graphics library.
- Understand basic keyboard interaction.
- Implement non-parameterized animation using fixed logic.

1.1 Code

```
#include <iostream>
#include <GL/glut.h>
#include <iostream>

const int WINDOW_WIDTH = 800;
const int WINDOW_HEIGHT = 600;

bool keyPressed[5] = { false, false, false, false, false };

void drawRect(int x, int y, int width, int height, float r, float g, float b) {
    glColor3f(r, g, b);
    glBegin(GL_QUADS);
    glVertex2i(x, y);
    glVertex2i(x + width, y);
    glVertex2i(x + width, y + height);
    glVertex2i(x, y + height);
    glEnd();
}

void drawScreenSplit() {

    glBegin(GL_POLYGON);
    glColor3f(0.2f, 0.2f, 0.2f);
    glVertex2i(0, 0);
    glVertex2i(300, 0);
    glVertex2i(300, 600);
    glVertex2i(0, 600);
    glEnd();

    glBegin(GL_POLYGON);
    glColor3f(0.9f, 0.9f, 0.9f);
    glVertex2i(300, 0);
    glVertex2i(800, 0);
    glVertex2i(800, 600);
    glVertex2i(300, 600);
    glEnd();
}
```

```

void drawSteeringWheel() {
    glColor3f(0.1f, 0.1f, 0.1f);
    glBegin(GL_POLYGON);
    glVertex2i(610, 300);
    glVertex2i(602, 330);
    glVertex2i(580, 352);
    glVertex2i(550, 360);
    glVertex2i(520, 352);
    glVertex2i(498, 330);
    glVertex2i(490, 300);
    glVertex2i(498, 270);
    glVertex2i(520, 248);
    glVertex2i(550, 240);
    glVertex2i(580, 248);
    glVertex2i(602, 270);
    glEnd();
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT);

    drawScreenSplit();

    drawRect(50, 470, 200, 40, 0.2f, 0.2f, 0.2f);
    glColor3f(1.0f, 1.0f, 1.0f);
    glRasterPos2i(70, 485);
    const char* leftLabel = "User Dashboard";
    for (int i = 0; leftLabel[i] != '\0'; i++)
        glutBitmapCharacter(GLUT_BITMAP_HELVETICA_18, leftLabel[i]);

    glColor3f(0.0f, 0.0f, 0.0f);
    glRasterPos2i(480, 410);
    const char* rightLabel = "Car Management System";
    for (int i = 0; rightLabel[i] != '\0'; i++)
        glutBitmapCharacter(GLUT_BITMAP_HELVETICA_18, rightLabel[i]);

    drawSteeringWheel();

    char keys[5] = { 'A', 'B', 'C', 'D', 'E' };
    for (int i = 0; i < 5; ++i) {

```

```

        int x = 300 + 100 + i * 70;
        int y = 100;
        drawRect(x, y, 60, 40,
            keyPressed[i] ? 0.0f : 0.5f,
            keyPressed[i] ? 1.0f : 0.5f,
            keyPressed[i] ? 0.0f : 0.5f);

        glColor3f(0.0f, 0.0f, 0.0f);
        glRasterPos2i(x + 20, y + 12);
        glutBitmapCharacter(GLUT_BITMAP_HELVETICA_18, keys[i]);
    }

    glutSwapBuffers();
}

void handleKey(unsigned char key, int x, int y) {
    if (key >= 'A' && key <= 'E') {
        int index = key - 'A';
        keyPressed[index] = !keyPressed[index];
        glutPostRedisplay();
    }
}

void init() {
    glMatrixMode(GL_PROJECTION);
    gluOrtho2D(0, WINDOW_WIDTH, 0, WINDOW_HEIGHT);
}

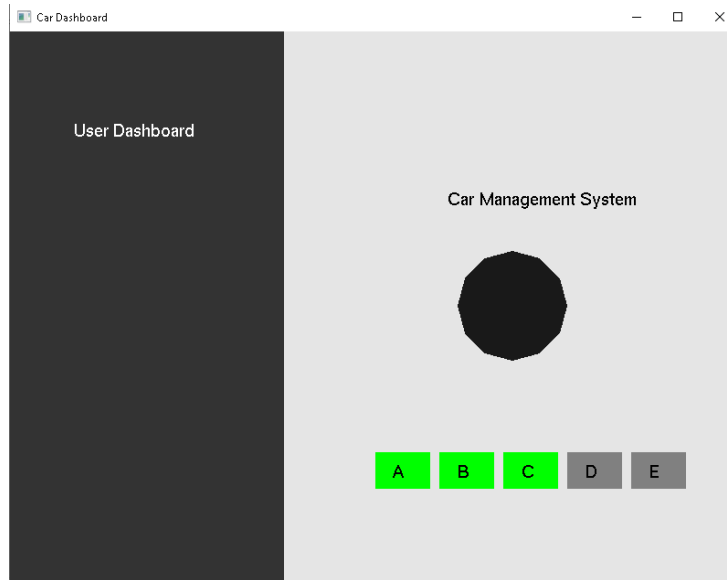
int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(WINDOW_WIDTH, WINDOW_HEIGHT);
    glutCreateWindow("Car Dashboard");

    init();
    glutDisplayFunc(display);
    glutKeyboardFunc(handleKey);
}

```

```
    glutMainLoop();  
    return 0;  
}
```

1.2 Output



1.3 Description

This OpenGL program simulates a simple 2D car dashboard interface using OpenGL and GLUT. The screen is divided into two sections: a left-side panel labeled "User Dashboard", and a right-side area displaying the "Car Management System". A stylized steering wheel is rendered in the center, and five interactive buttons labeled A to E are displayed below it. Users can press the corresponding keys (A to E) on their keyboard to toggle the button colors, visually indicating their activation status. The project demonstrates basic 2D graphics, screen layout, and keyboard input handling using GLUT.
